

The Title of A Simple MIT Press Style

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Abstract

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Să se determine o formulă de cuadratură cu grad maxim de exactitate de forma:

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = Af(-1) + Bf(x_0) + Cf(1) + R(f)$$

Cf. curs nu e de tip Gauss (nodurile ar fi sol. ec.: $\cos(3 \arccos x) = 0$, care nu conțin ± 1). Punem cond. ca $R(f) = 0$ pentru f polinom de grad cât mai mare:

$$R(1) = 0$$

$$R(x) = 0$$

$$R(x^2) = 0$$

$$R(x^3) = 0.$$

Rezultă sistemul:

$$\begin{cases} \int_{-1}^1 \frac{1}{\sqrt{1-x^2}} dx - A - B - C = 0 \\ \int_{-1}^1 \frac{x}{\sqrt{1-x^2}} dx - A(-1) - Bx_0 - C = 0 \\ \int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx - A(-1)^2 - Bx_0^2 - C1^2 = 0 \\ \int_{-1}^1 \frac{x^3}{\sqrt{1-x^2}} dx - A(-1)^3 - Bx_0^3 - C1^3 = 0 \end{cases}$$

: π

$$\int_{-1}^1 \frac{x^2}{\sqrt{1-x^2}} dx \stackrel{x=\sin t}{=} \int_{-\pi/2}^{\pi/2} \frac{\sin^2 t}{\cos t} \cos t dt = \int_{-\pi/2}^{\pi/2} \frac{1-\cos 2t}{2} dt = \frac{\pi}{2}$$

Sistemul:

$$\begin{cases} A + B + C = \pi \\ -A + Bx_0 + C = 0 \\ A + Bx_0^2 + C = \frac{\pi}{2} \\ -A + Bx_0^3 + C = 0 \end{cases}$$

Rezultă:

$$B(x_0^3 - x_0) = 0$$

deci:

$$x_0 = 0.$$

$$A = C; A + C = \frac{\pi}{2}$$

adică:

$$A = C = \frac{\pi}{4}$$

$$B = \frac{\pi}{2}$$

Formula de cuadratură:

$$\int_{-1}^1 \frac{f(x)}{\sqrt{1-x^2}} dx = \frac{\pi}{4} (f(-1) + 2f(0) + f(1)) + R(f)$$

$$R(x^4) = \int_{-1}^1 \frac{x^4}{\sqrt{1-x^2}} dx - \frac{\pi}{4} (1 + 1) = -\frac{\pi}{8}$$

$$R(f) = -\frac{\pi}{8} \frac{f^{iv}(\xi)}{4!}$$